10397
Connect the Campus

110030022a 曾廷綸

題目大意

學校要在各大樓間增設電纜，在不能改動線有電纜的條件下，希望能以最少費用安裝完成。

INPUT		/OUTPUT
4	大樓數量	4.41
103 104	大樓座標	還需要的電纜長
104 100		
104 103		
100 100		
1	已有電纜數量	
4 2	已連接的大樓	

＊電纜之間沒有方向性

＊建築物編號從１開始

＊每對建築物只需要一根電纜連接

＊沒有兩個建築物在同一個座標點

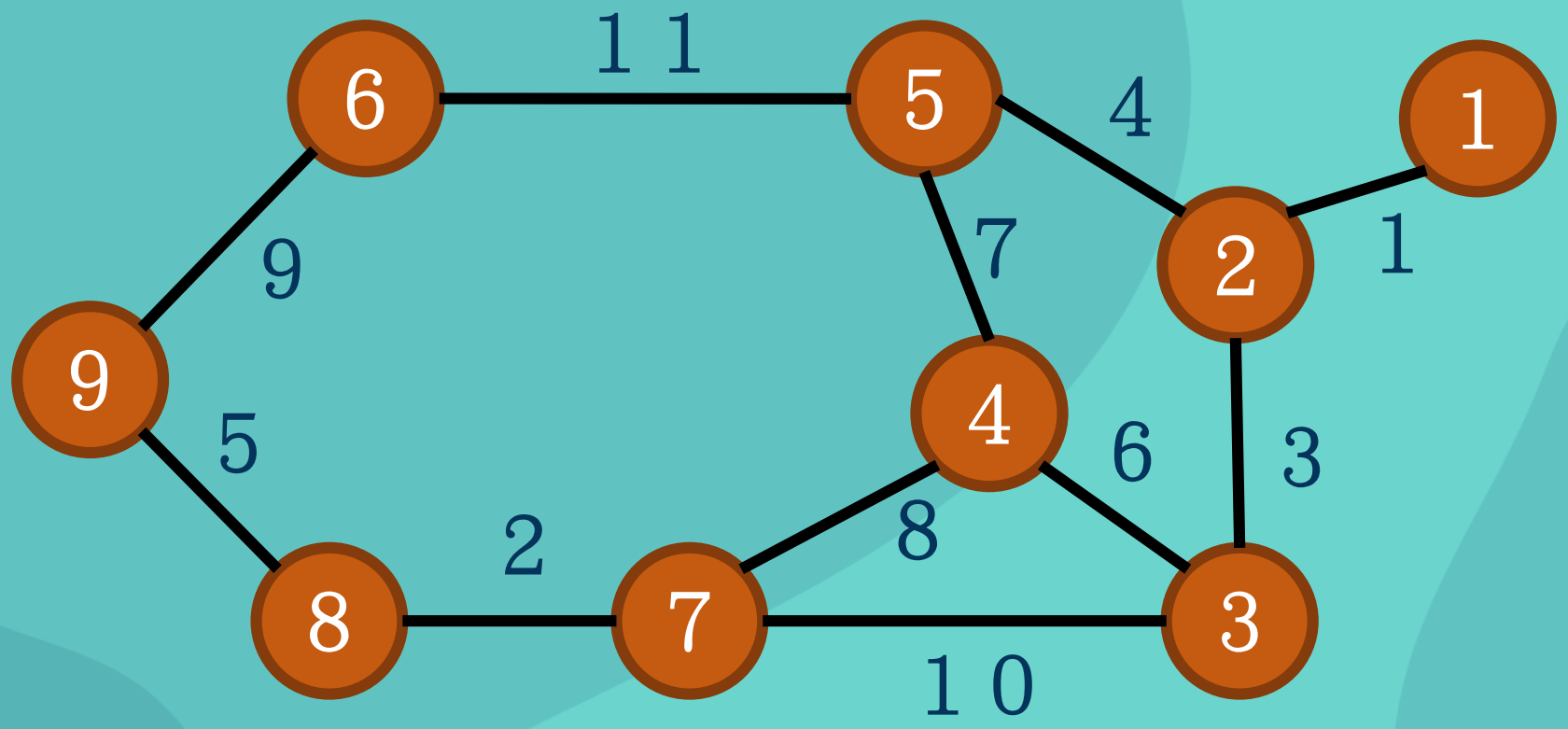
Minimum Spanning Tree (greedy)

1. 新建圖，圖中擁有原圖中相同的節點，但沒有邊
2. 在原圖中加入所有可能邊，使其成為complete graph，並將原圖中所有的邊按權重從小到大排序
3. 從權重最小的邊開始，如果這條邊連接的兩個節點於圖G中不在同一個連通塊中，則添加這條邊到圖$\mathscr{G}$中
4. 重複3，直至圖$\mathscr{G}$中所有的節點都在同一個連通分量中

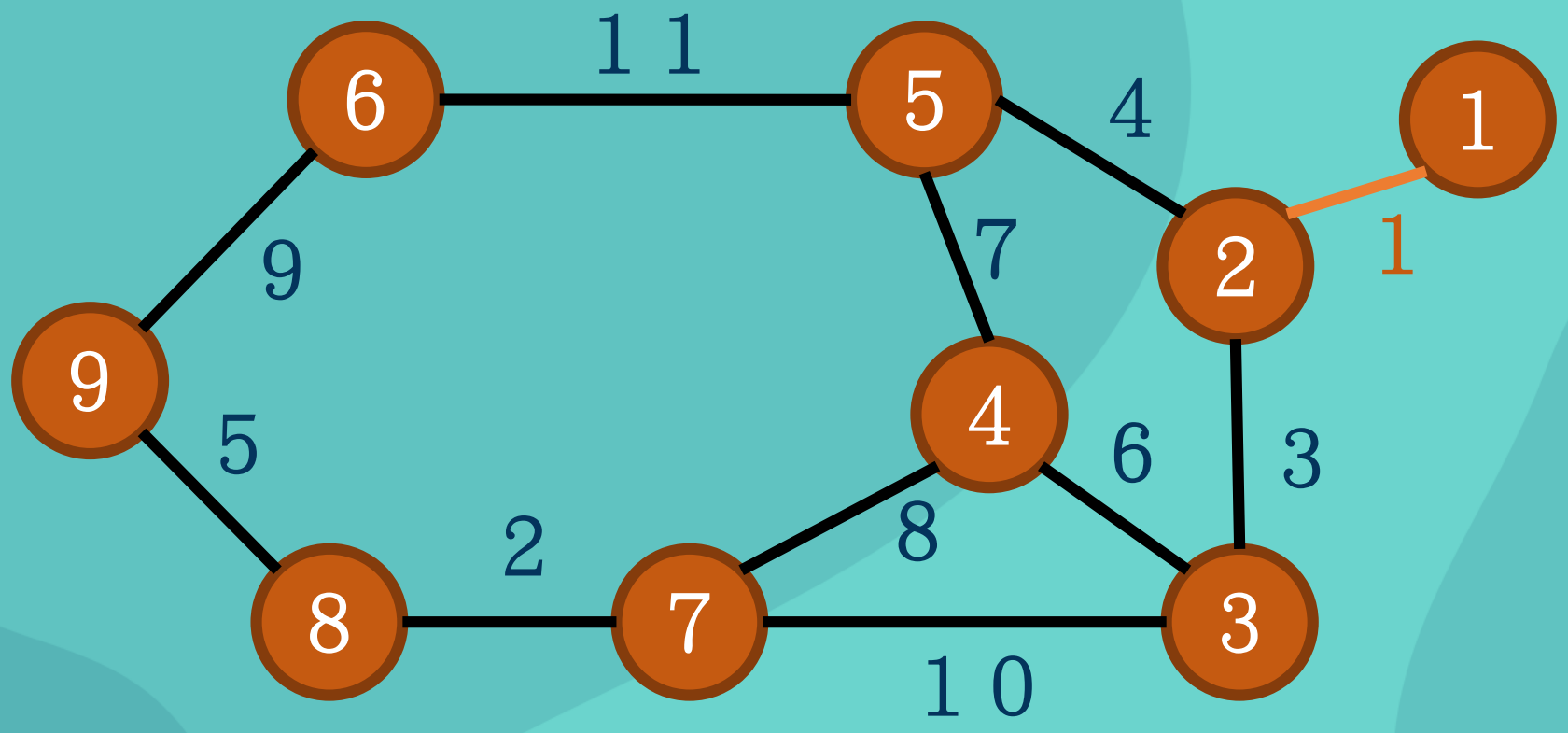
*連通塊：擁有不同的 Ancestor

Minimum Spanning Tree (greedy)

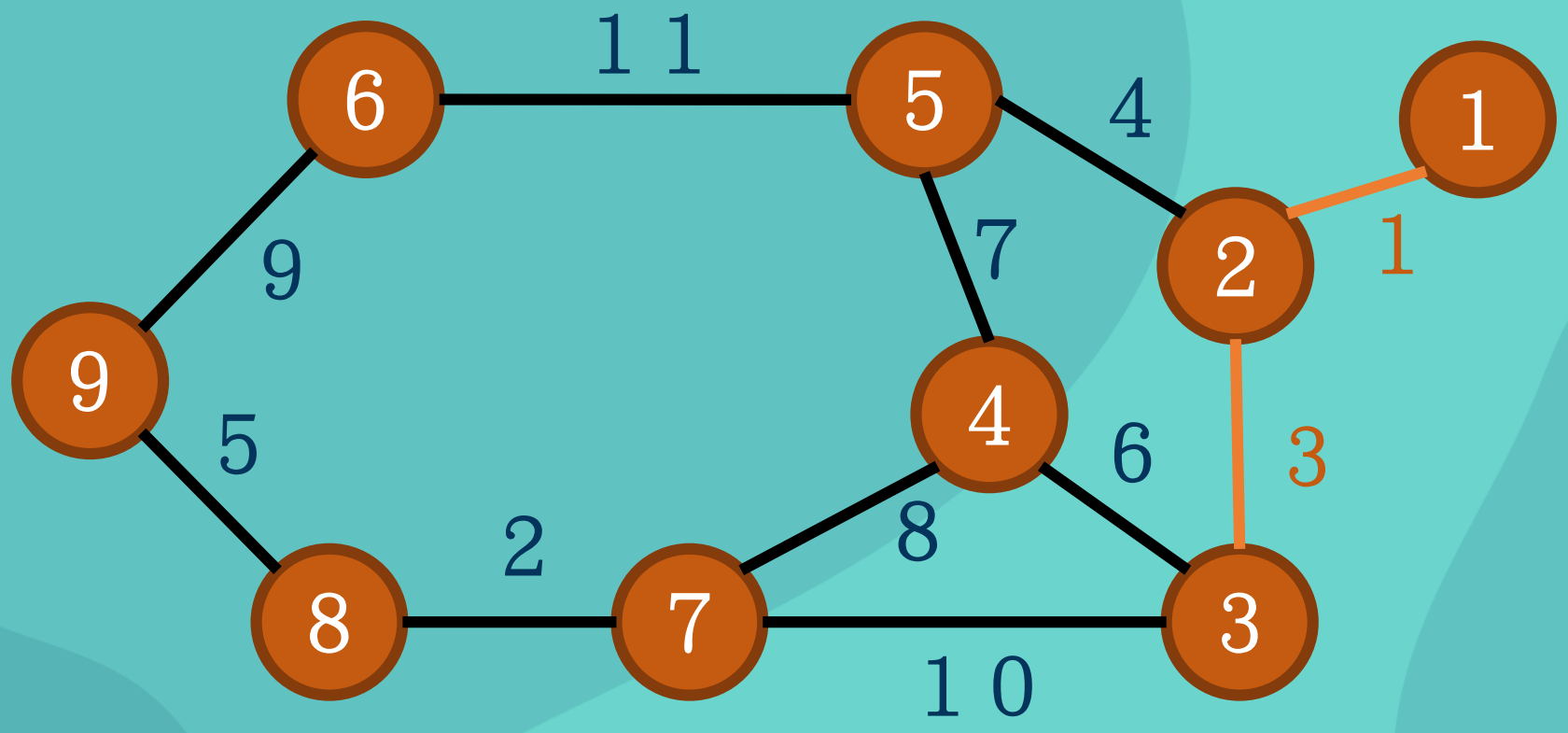
＊白數字為建築物編號
＊黑數字為邊長排序
（以2為起點）



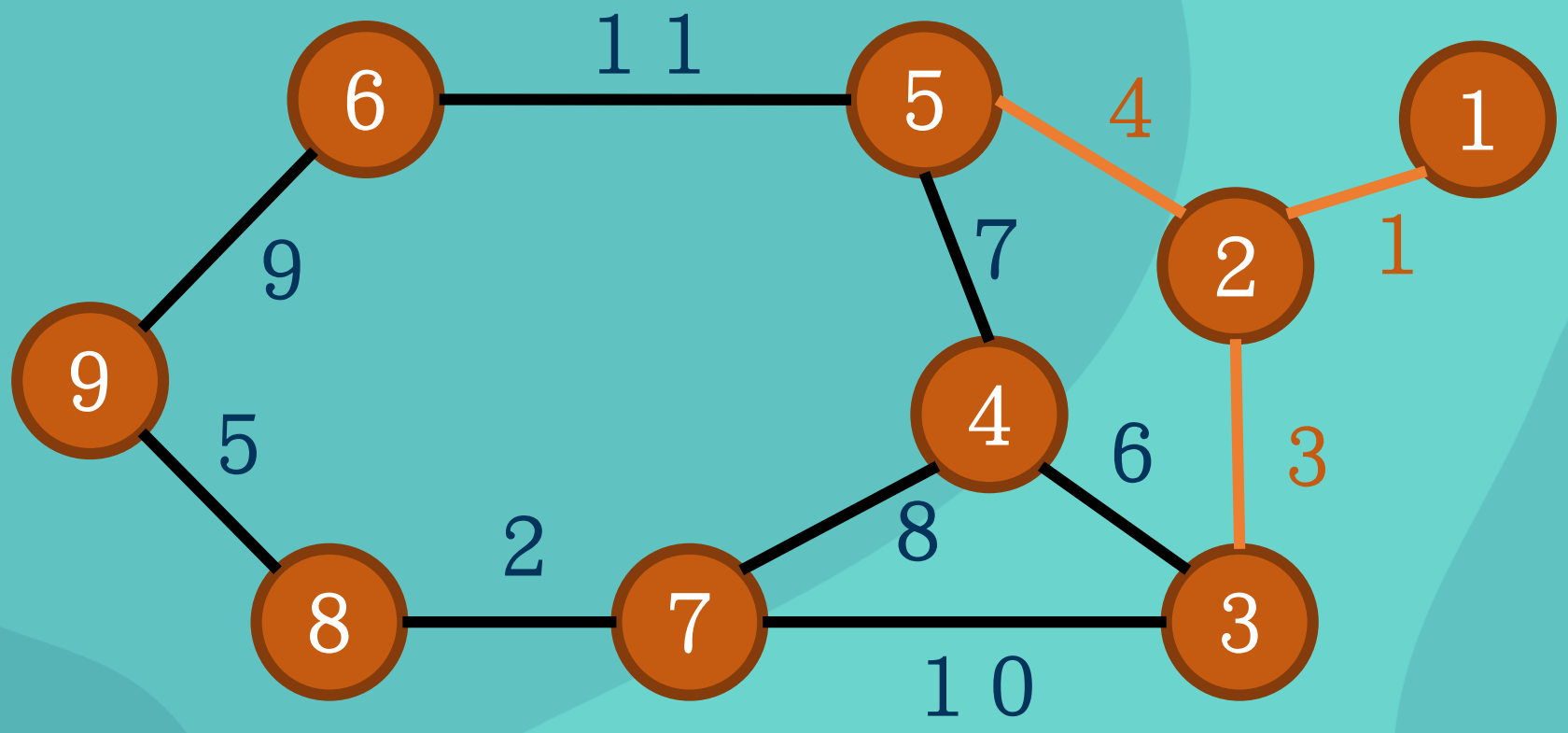
Minimum Spanning Tree (greedy)



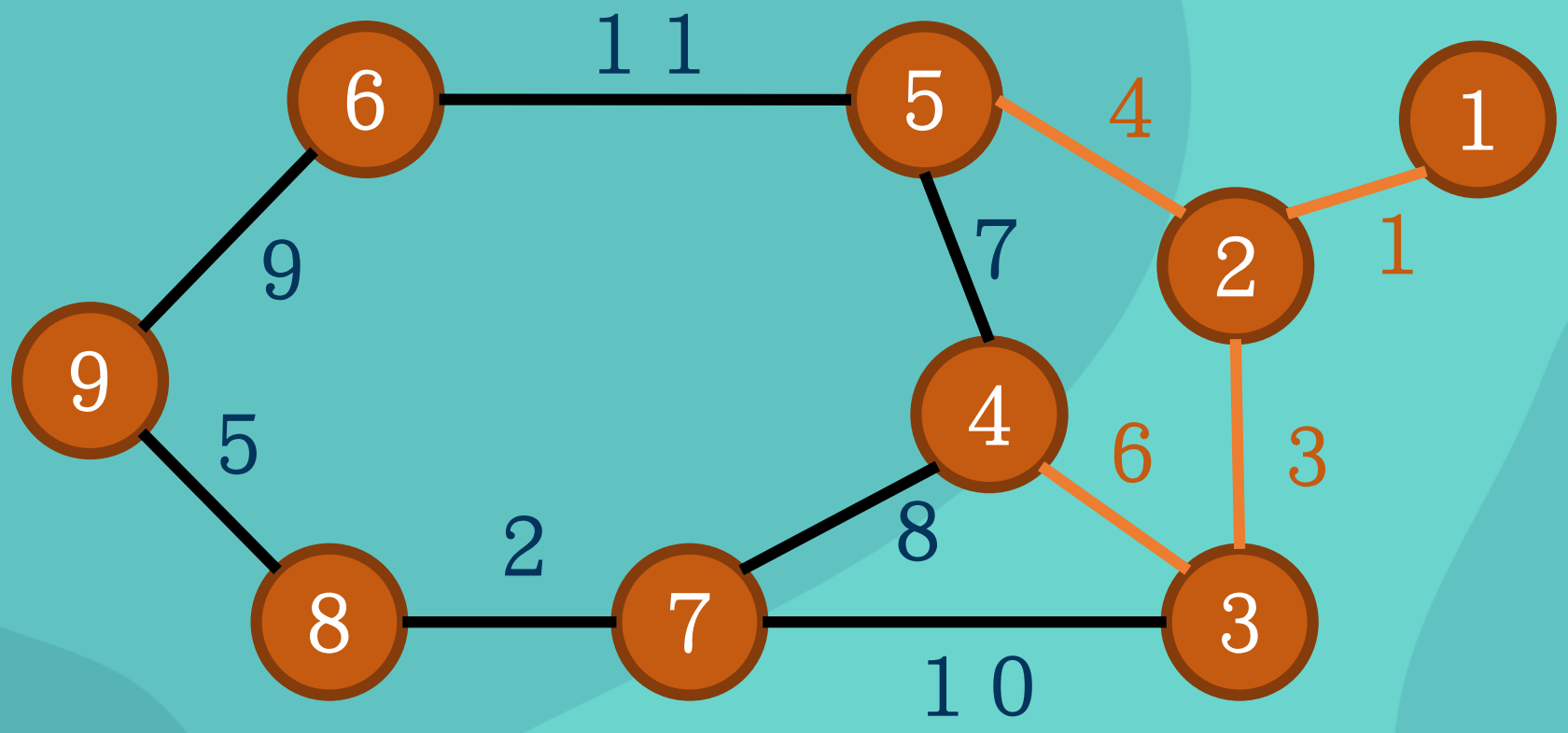
Minimum Spanning Tree (greedy)



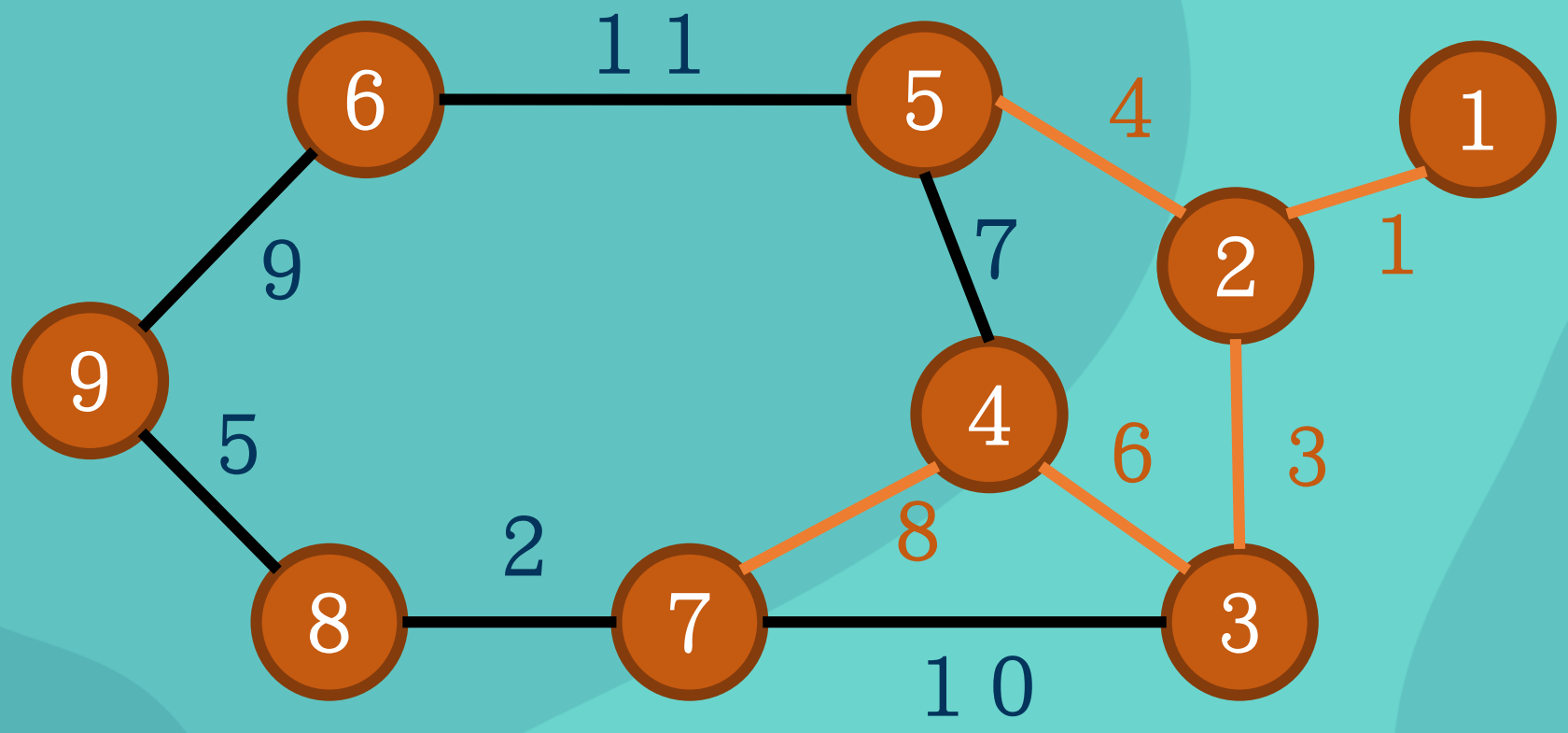
Minimum Spanning Tree (greedy)



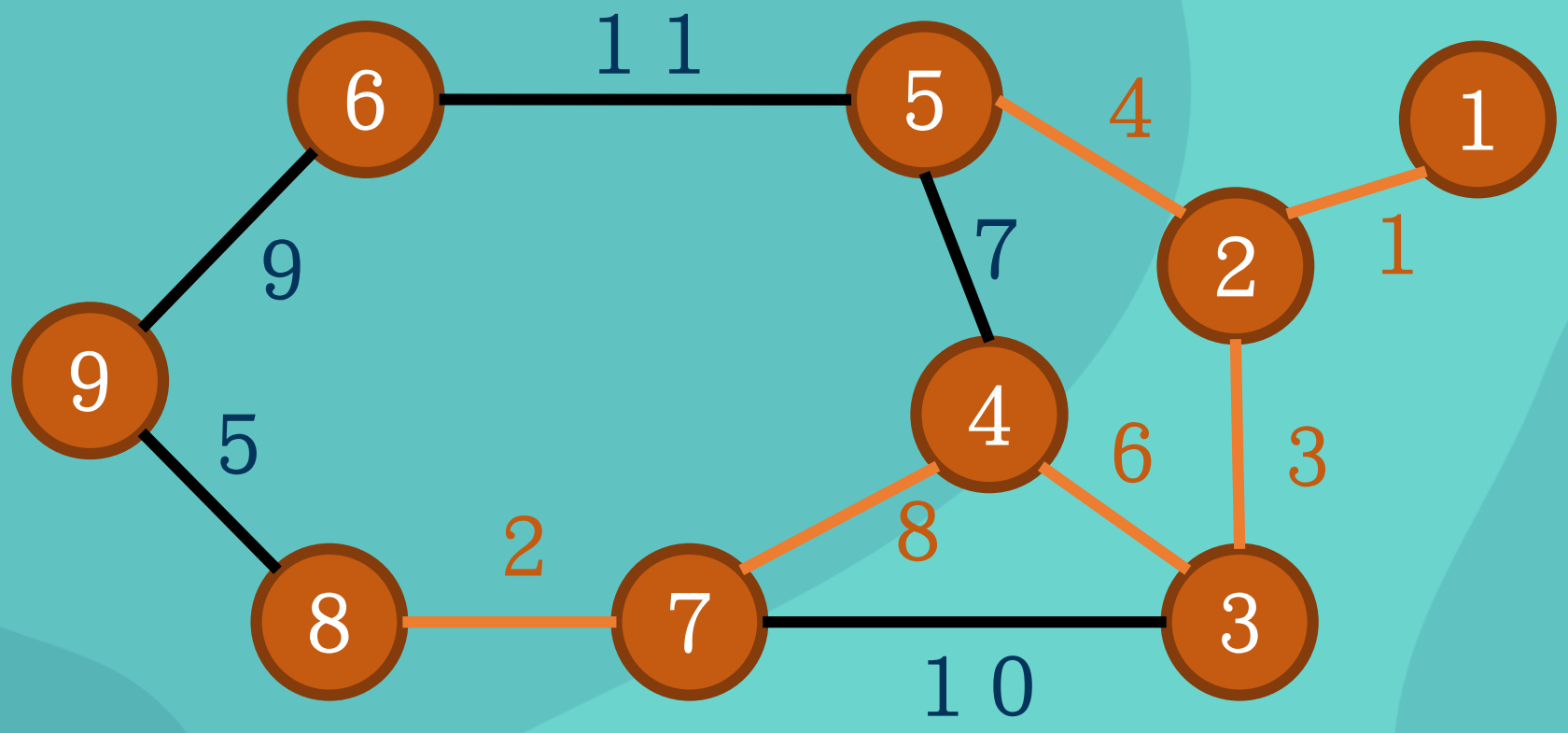
Minimum Spanning Tree (greedy)



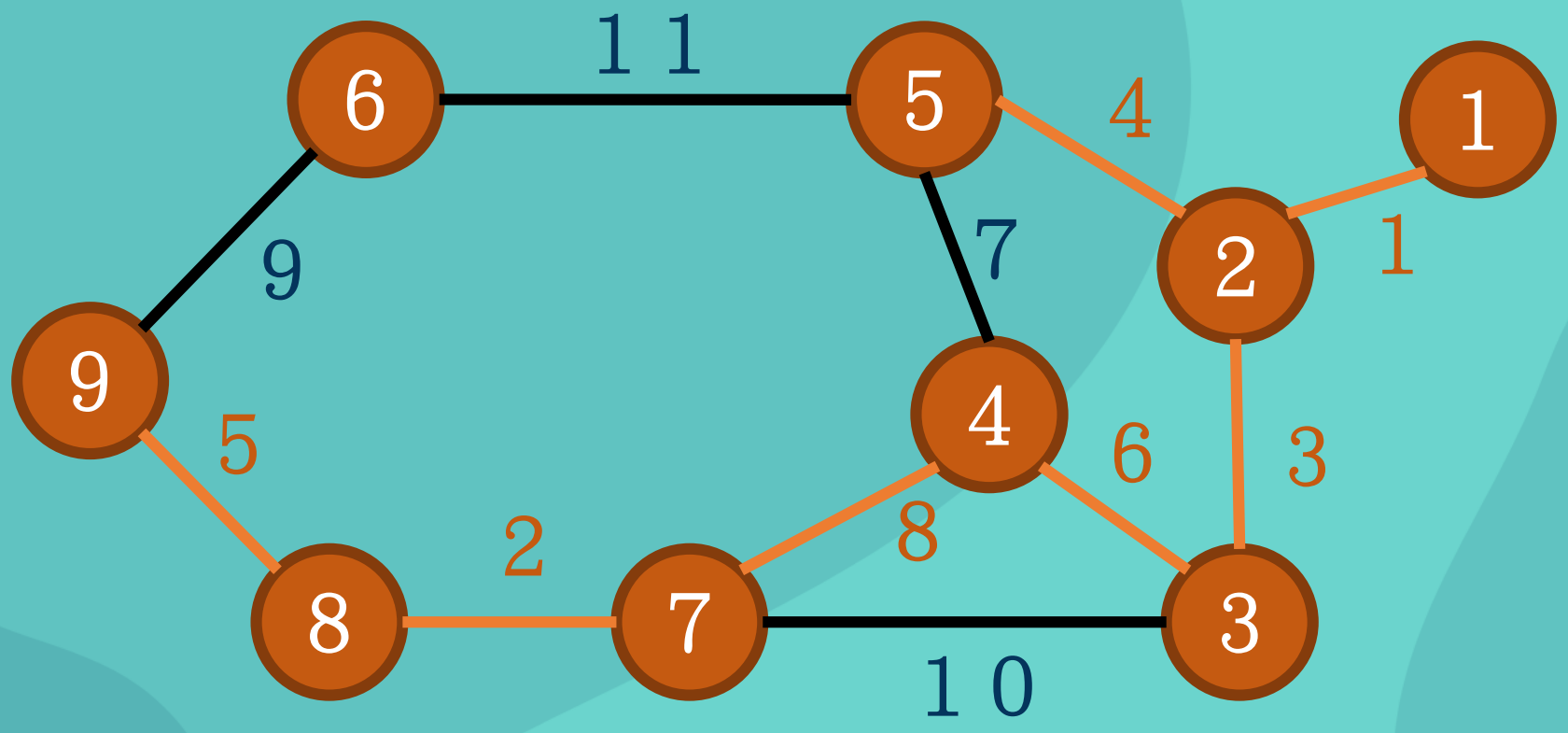
Minimum Spanning Tree (greedy)



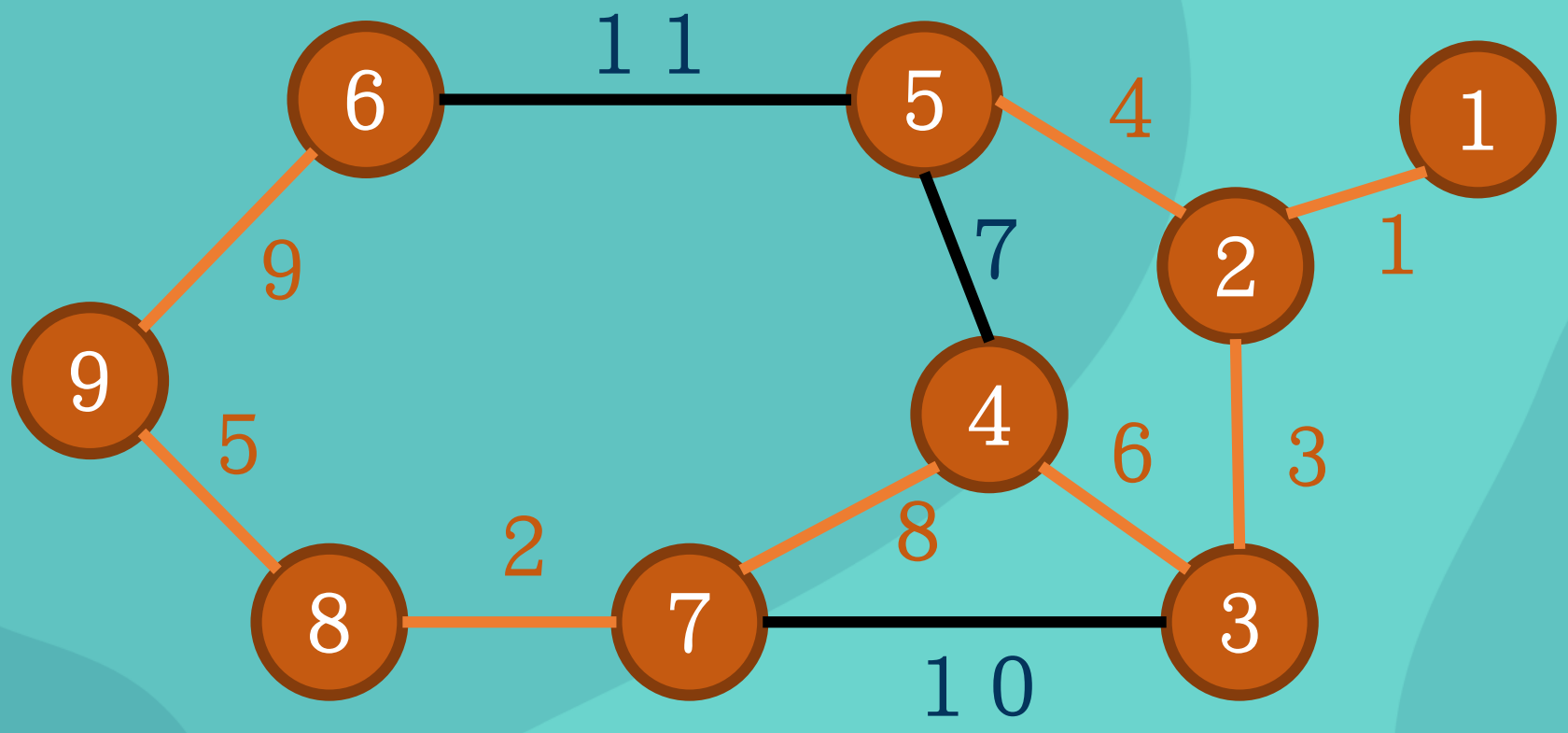
Minimum Spanning Tree (greedy)

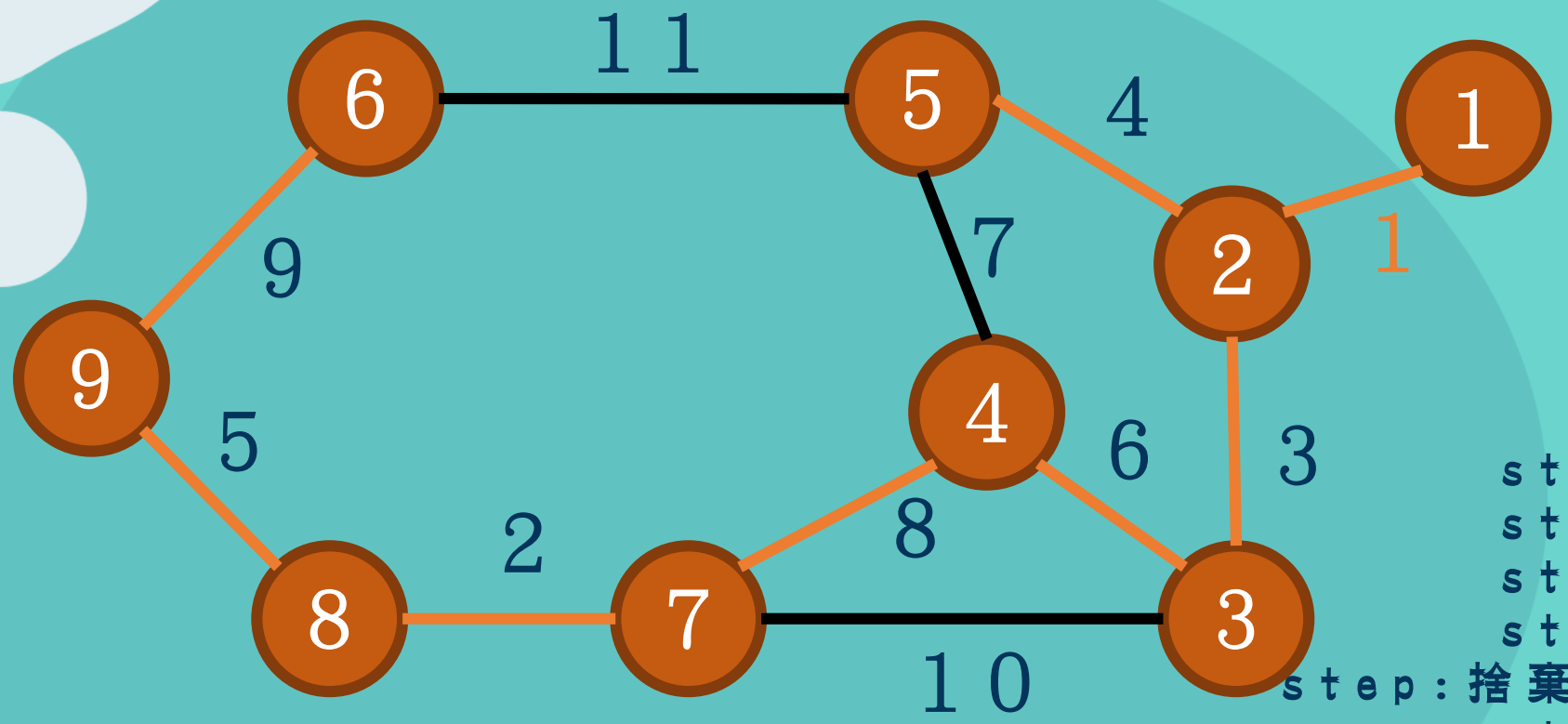


Minimum Spanning Tree (greedy)



Minimum Spanning Tree (greedy)





總整理

step：選擇（1-2）
step：選擇（2-3）
step：選擇（2-5）
step：選擇（3-4）
step：捨棄（4-5）（會產生cycle）
step：選擇（4-7）
step：選擇（7-8）
step：選擇（8-9）
step：選擇（6-9）
step：捨棄（3-7）（會產生cycle）
step：捨棄（5-6）（會產生cycle）

Minimum Spanning Tree (greedy)

```
Procedure PrimsMST(Graph):
Vnew[] = {x}
Enew[] = {}
while Vnew is not equal to V
    u -> a node from Vnew
    v -> a node that is not in Vnew such that edge u-v has the minimum cost

    add v to Vnew
    add edge (u, v) to Enew
end while
Return Vnew and Enew
```